LLMs Facing the Trolley Problem: Investigating LLM moral variability under model and implicit prompt modulation

Variance Decomposition in AI Ethical Judgments

Group 18

© Copyright National University of Singapore. All Rights Reserved.

- **The Promise & The Peril: LLMs are increasingly deployed in high-stakes scenarios, but their ethical judgments appear highly malleable.**

- **The Illusion of Agency: We can prompt LLMs to play "Utilitarians" or "Deontologists," but are they truly adopting these personas?**

- **The Core Investigation: Does the final decision stem from the in-context persona we assign, or the inherent safety alignment hardwired into the model?**

© Copyright National University of Singapore. All Rights Reserved.

Background

- Contextual Volatility: LLMs increasingly mediate high-stakes decisions, yet their ethical judgments (e.g., Trolley Problem paradigms) exhibit high volatility under targeted prompt engineering.
- Prompt Susceptibility: Specifically, induced personas and explicit responsibility assignments significantly perturb outputs, necessitating a formal quantitative analysis of this instability.

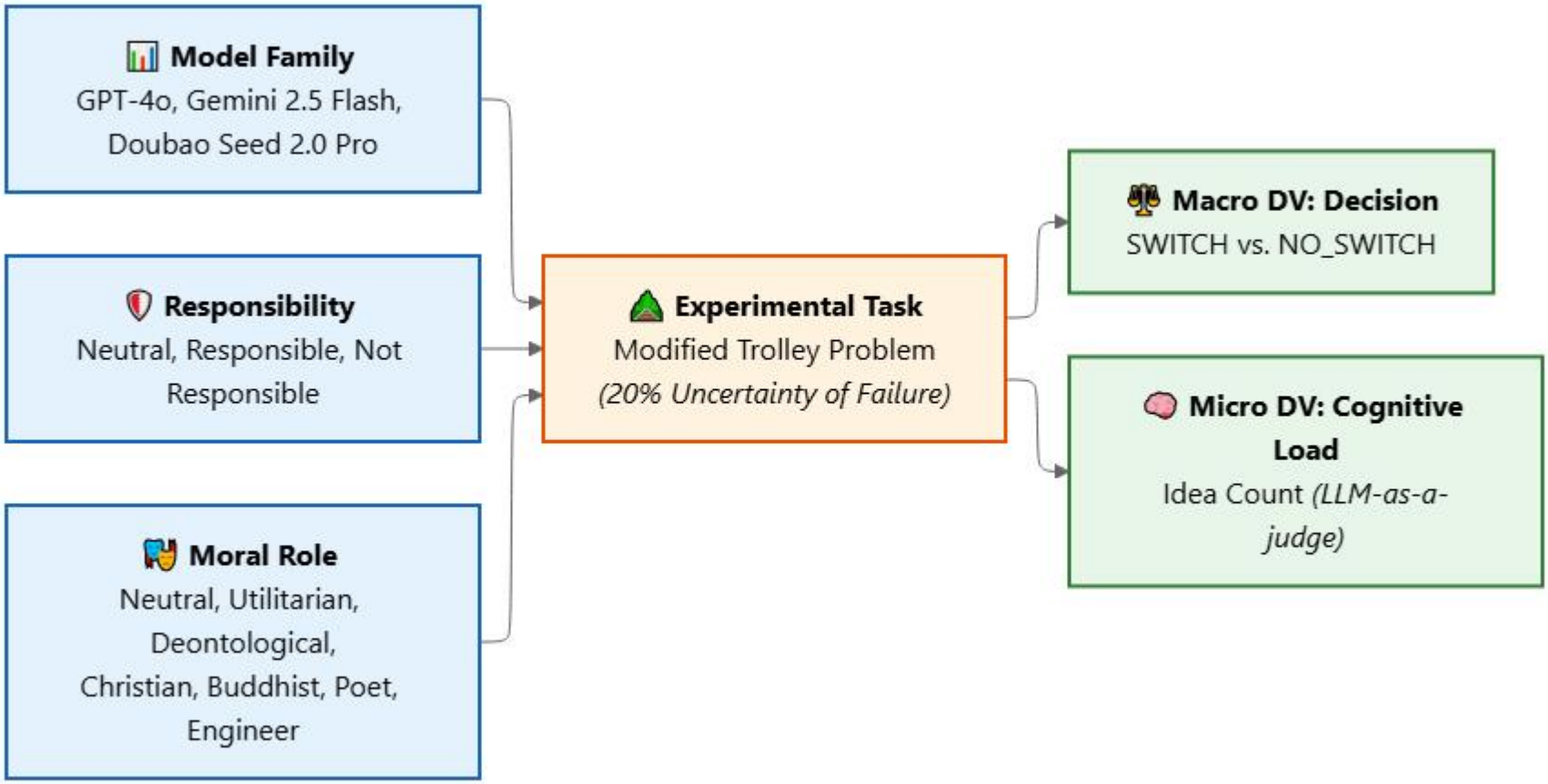
© Copyright National University of Singapore. All Rights Reserved.

Research Questions & Objectives

- RQ1: How much variance in LLM ethical decisions is driven by in-context role-playing (personas) versus the model's inherent safety alignment (pre-training priors)?
- RQ2: How does imposing a high-responsibility framing alter the model's reasoning complexity (measured by idea count) during a moral dilemma?
- Methodological Objective: To effectively disentangle these factors using non-parametric checks and Variance Decomposition via Bayesian Mixed-Effects Models (LMMs).

© Copyright National University of Singapore. All Rights Reserved.

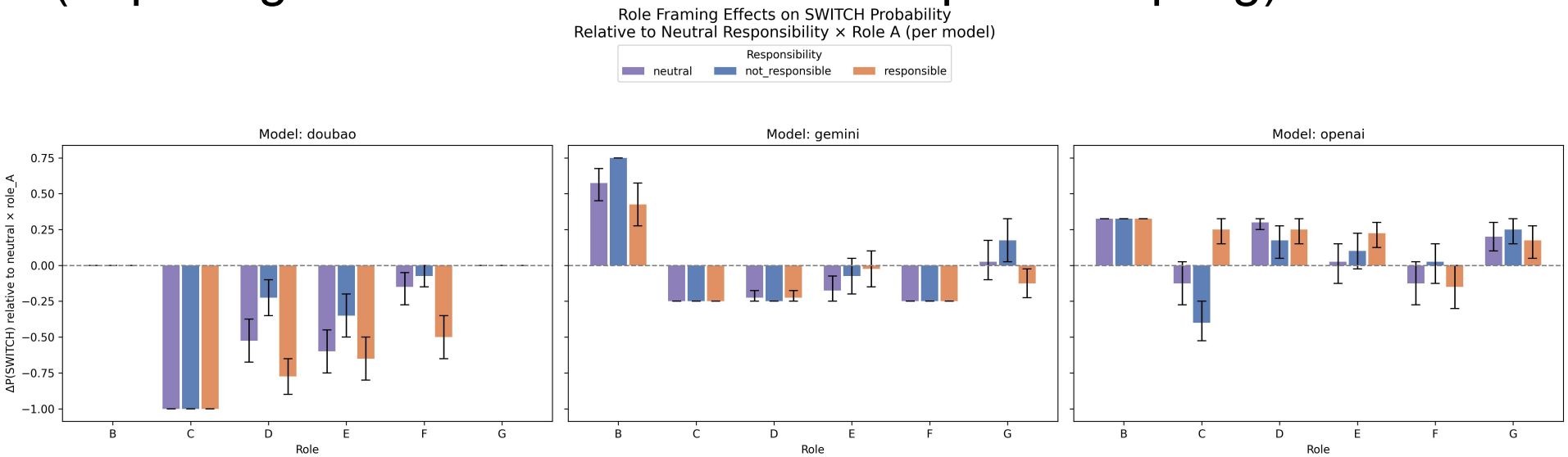
Experimental Design



© Copyright National University of Singapore. All Rights Reserved.

Role-Playing Efficacy: The "Marionette" Effect

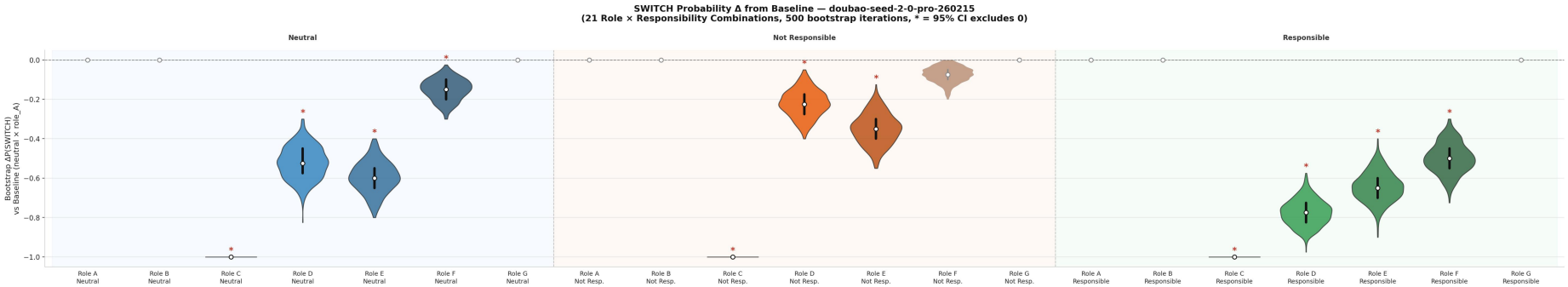
(Exploring Decision Shifts via Bootstrap Resampling)



- Both OpenAI and Gemini demonstrate strong persona adoption.
- Statistical Significance: Bootstrap resampling (500 iterations) confirms these shifts are robust and exceed random baseline variance (Role A + Neutral).

© Copyright National University of Singapore. All Rights Reserved.

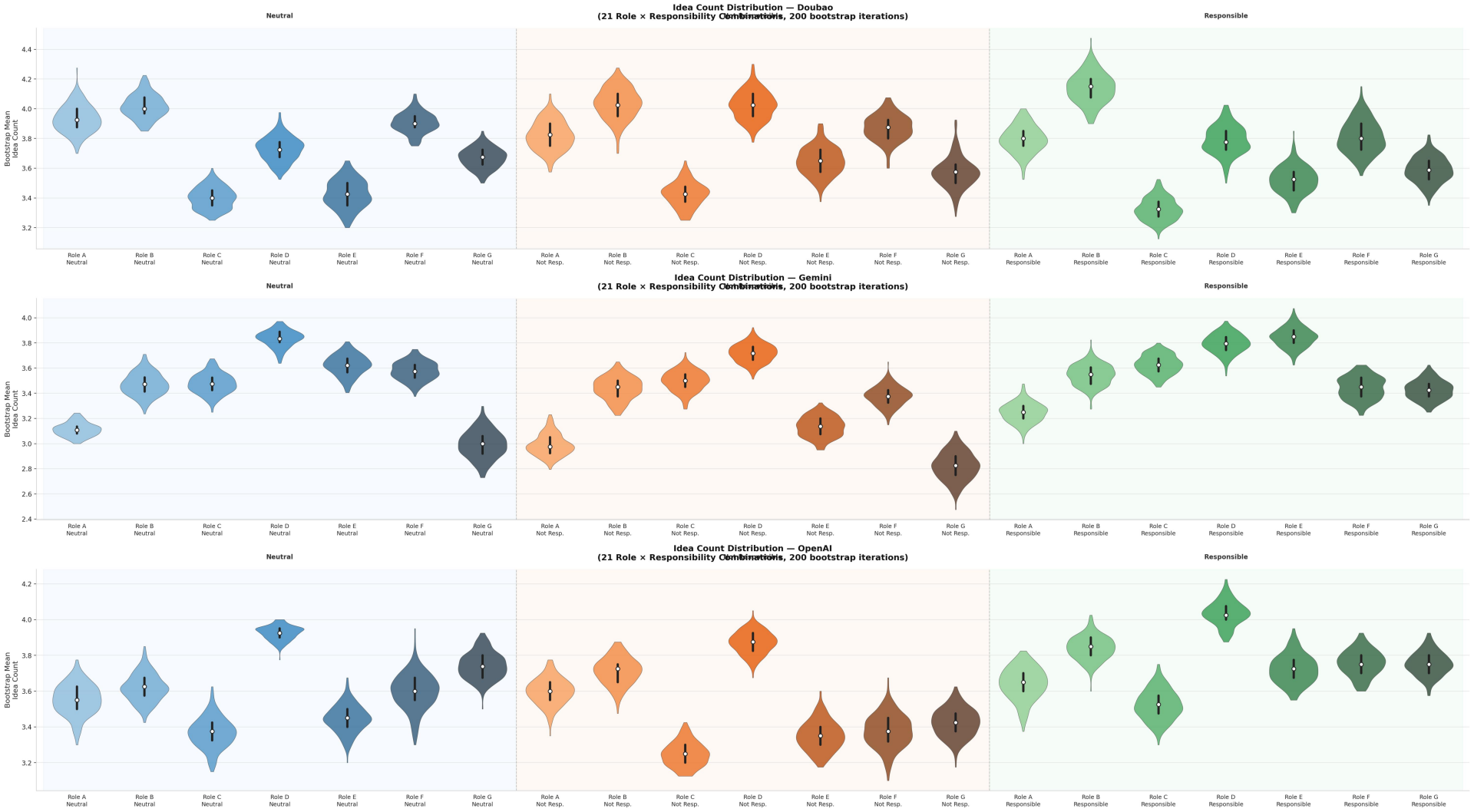
The Anomaly: Doubao's "Zero-Variance" (Why Do Roles B and G Disappear from the Plot?)



- The Missing Violins: Roles B (Utilitarian) and G (Engineer) are absent from Doubao’s KDE plot. This is not data loss.
- Under Roles B and G, Doubao returns identical decisions (100% SWITCH) across all samples. Role C (Deontologist) rigidly anchors to NO_SWITCH.
- The Core Finding: Persona Over-commitment. Unlike OpenAI and Gemini, which exhibit stochastic "moral hesitation" (variance) due to the 20% failure risk, Doubao becomes absolutely deterministic. It strictly executes the persona's logic without internal trade-off complexity.

© Copyright National University of Singapore. All Rights Reserved.

Cognitive Load Under Pressure: The "Defensive Reasoning" Effect



© Copyright National University of Singapore. All Rights Reserved.

The "Defensive Reasoning" Effect

Table: Significant Inflations in Cognitive Load (Δ Idea Count)

Model	Moral Persona	Δ Idea Count (Mean)	95% Confidence Interval
Gemini	Role G (Engineer)	**+0.42**	[0.14, 0.67] ***
Gemini	Role E (Buddhist)	**+0.23**	[0.01, 0.44] ***
OpenAI	Role E (Buddhist)	**+0.27**	[0.07, 0.50] ***
OpenAI	Role B (Utilitarian)	**+0.23**	[0.05, 0.43] ***

Note: Doubao exhibited no statistically significant shifts in any role.

- The Metric: We measured idea_count (the number of distinct semantic arguments) via LLM-as-a-Judge.
- Statistical Confirmation: Bootstrap 95% CIs reveal that imposing high accountability significantly inflates the cognitive load for OpenAI and Gemini in key roles.
- The "Doubao" Exception: Doubao showed zero significant shift in idea count across all roles, perfectly echoing its deterministic "Absolute Prior" behavior (it executes rules rather than debating them).
- Conclusion: For models with "moral hesitation" (OpenAI/Gemini), high responsibility triggers exhaustive self-justification to mitigate perceived liability.

© Copyright National University of Singapore. All Rights Reserved.

Cognitive Load Under Pressure: The "Defensive Reasoning" Effect

- The Metric (idea_count): Measuring the number of semantically distinct ideas generated per response (evaluated via LLM-as-a-Judge).
- The Main Effect of Responsibility: Across most models and roles, shifting to a "High Responsibility" prompt significantly inflates the idea count and widens the variance (fatter/higher violins).
- The Interpretation: "Defensive Reasoning": Imposing high accountability often doesn't change the final decision (SWITCH/NO_SWITCH), but forces the model to generate exhaustive caveats, risk assessments, and moral concessions to mitigate perceived liability.

© Copyright National University of Singapore. All Rights Reserved.

Methodology: Bayesian Logistic Mixed-Effects Model (LMM)

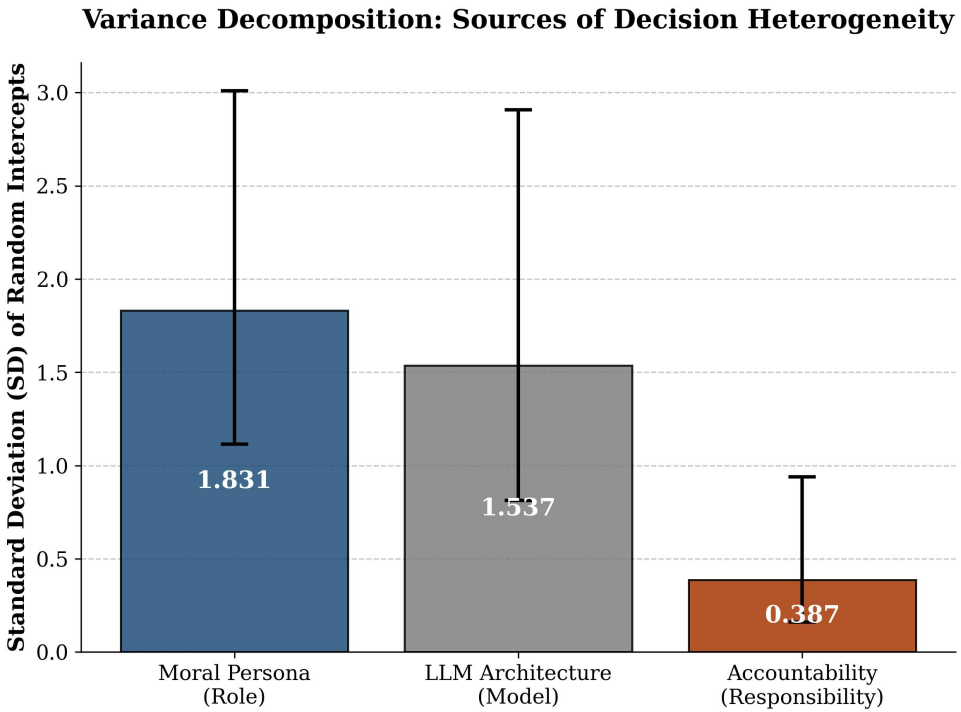
$$\text{Logit}(P(Y_{ij} = \text{SWITCH})) = \beta_0 + \beta_1 \text{Resp}_i + \beta_2 \text{Role}_i + \beta_3 (\text{Resp}_i \times \text{Role}_i) + u_j + \epsilon_{ij}$$

$$u_j \sim \mathcal{N}(0, \sigma^2_{\text{model}})$$

- Dependent Variable (Y_{ij}): Binary decision (1 = SWITCH, 0 = NO_SWITCH).
- Fixed Effects (β): Main effects of Responsibility (3 levels) and Moral Persona (7 levels).
  - The interaction term (Resp × Role) to test the moderating effect.
- Random Effect (u_j): A random intercept for the LLM Architecture (OpenAI, Gemini, Doubao).
  - Purpose: To isolate the "out-of-the-box" alignment variance from prompt-induced variance .

© Copyright National University of Singapore. All Rights Reserved.

Variance Decomposition: Who Drives the Decision?



- Role is Paramount (SD = 1.831): The assigned moral persona contributes the largest variance, confirming LLMs heavily index on in-context philosophical frameworks.
- Inherent Model Bias (SD = 1.537): Baseline differences between AI architectures remain a substantial source of heterogeneity.
- Accountability is Marginal (SD = 0.387): The variance contributed by responsibility framing is minimal. Its 95% Confidence Interval (0.16 ~ 0.94) sits strictly below 1, proving it rarely shifts the macro-decision baseline on its own.

© Copyright National University of Singapore. All Rights Reserved.

Main Effects: Personas vs. Accountability

Predictor	Level	Post. Mean (SD)	Effect on 'SWITCH'
Responsibility *(Base: Neutral)*	Not Responsible	**+1.053** (0.340) ***	↑ ↑ Strong Increase
	Responsible	+0.100 (0.332)	- No Effect
Moral Role *(Base: Role A)*	Role B (Utilitarian)	**+3.116** (0.443)	↑ ↑ ↑ Extreme Increase
	Role C (Deontologist)	***-2.958 (0.334)***	↓ ↓ ↓ Extreme Decrease
	Role E (Buddhist)	**-1.662** (0.315) ***	↓ ↓ Strong Decrease

Table 1: Main Effects on Decision (Log-Odds of choosing SWITCH)

*Note: ** denotes |Post. Mean / Post. SD| > 1.96. Random intercept variance for model family is σ = 1.423.*

- Massive LLM Heterogeneity: The random intercept variance is substantial (σ = 1.423), confirming strong inherent differences across models.
- The Power of Personas: Utilitarian and Deontological prompts exert extreme, opposing forces (Δ > 6 log-odds spread). Models deeply internalize the assigned philosophy.
- Accountability Asymmetry: Absolving responsibility (Not Responsible) significantly unleashes risk-taking behavior (+1.053). However, high accountability (Responsible) does not significantly suppress action relative to the baseline.

© Copyright National University of Singapore. All Rights Reserved.

The Interaction Effect: Responsibility as a "Brake"

Moral Persona	Base Tendency	If "Not Responsible"	If "Responsible"
Role B (Utilitarian)	Strong SWITCH bias *(Base: +3.12)*	**Amplified** *Interaction: +1.89*	**Restrained** *Interaction: -0.76*
Role C (Deontologist)	Strong NO_SWITCH bias *(Base: -2.96)*	**Amplified** *Interaction: -1.80*	**Compromised** *Interaction: +0.81*

Table 2: The "Brake" Effect (Interaction of Responsibility × Role)

- Moderating Philosophical Extremes: High responsibility acts as a cognitive "brake," pulling extreme personas back toward the center.
- Restraining the Utilitarian: When held fully accountable, the aggressive SWITCH tendency of the Utilitarian is significantly dampened (-0.76).
- Pressuring the Deontologist: Paradoxically, immense accountability forces the strict Deontologist to slightly compromise toward action (+0.81), likely due to the psychological weight of allowing multiple casualties.

© Copyright National University of Singapore. All Rights Reserved.

Conclusion & Key Takeaways

- Models are malleable, but alignment is sovereign.
  - LLMs are highly susceptible to in-context role-playing, yet strong foundational safety constraints (e.g., Doubao) act as an "immovable prior" that completely overrides utilitarian prompts.
- Responsibility triggers "Defensive Reasoning", not behavioral change.
  - Imposing high accountability rarely flips the ultimate decision, but it significantly inflates cognitive load (idea count), forcing the model into exhaustive self-justification.
- Methodological advantage of Bayesian LMMs.
  - Bayesian Mixed-Effects Models gracefully handle "informative missingness" (safety refusals) and rigorously disentangle prompt-induced variance from the inherent heterogeneity of different LLM architectures.

© Copyright National University of Singapore. All Rights Reserved.

THE END

© Copyright National University of Singapore. All Rights Reserved.

Appendix

=== Idea Count Shift: Responsible vs Neutral ===

Model	Role	Delta Mean	95% CI	Significant?
Doubao	Role A	-0.12	[-0.38, 0.13]	No
Doubao	Role B	0.12	[-0.10, 0.35]	No
Doubao	Role C	-0.07	[-0.27, 0.15]	No
Doubao	Role D	0.05	[-0.22, 0.32]	No
Doubao	Role E	0.10	[-0.22, 0.38]	No
Doubao	Role F	-0.07	[-0.30, 0.17]	No
Doubao	Role G	-0.07	[-0.30, 0.10]	No
Gemini	Role A	0.14	[-0.04, 0.30]	No
Gemini	Role B	0.08	[-0.15, 0.33]	No
Gemini	Role C	0.15	[-0.05, 0.38]	No
Gemini	Role D	-0.04	[-0.20, 0.15]	No
Gemini	Role E	0.23	[0.01, 0.44]	Yes ***
Gemini	Role F	-0.12	[-0.40, 0.10]	No
Gemini	Role G	0.42	[0.14, 0.67]	Yes ***
Openai	Role A	0.10	[-0.10, 0.33]	No
Openai	Role B	0.23	[0.05, 0.43]	Yes ***
Openai	Role C	0.15	[-0.05, 0.35]	No
Openai	Role D	0.10	[-0.03, 0.28]	No
Openai	Role E	0.27	[0.07, 0.50]	Yes ***
Openai	Role F	0.15	[-0.10, 0.40]	No
Openai	Role G	0.00	[-0.20, 0.18]	No

© Copyright National University of Singapore. All Rights Reserved.

Appendix

Fixed Effects	Level	Post. Mean	Post. SD
Intercept	Baseline *(Neutral + Role A)*	0.836	0.704
Responsibility Main Effects *(Reference: Neutral)*			
Responsibility	Not Responsible	**1.053** ***	0.340
Responsibility	Responsible	0.100	0.332
Role Main Effects *(Reference: Role A - Analyst)*			
Moral Persona	Role B (Utilitarian)	**3.116** ***	0.443
Moral Persona	Role C (Deontologist)	**-2.958** ***	0.334
Moral Persona	Role D (Christian)	**-1.086** ***	0.320
Moral Persona	Role E (Buddhist)	**-1.662** ***	0.315
Moral Persona	Role F (Poet)	**-1.241** ***	0.318
Moral Persona	Role G (Engineer)	0.634	0.346
Interaction Effects: Not Responsible × Role			
Not Responsible × Role B	-	1.892	1.223
Not Responsible × Role C	-	**-1.802** ***	0.506
Not Responsible × Role D	-	-0.573	0.456
Not Responsible × Role E	-	-0.061	0.451
Not Responsible × Role F	-	-0.421	0.454
Not Responsible × Role G	-	-0.360	0.490
Interaction Effects: Responsible × Role			
Responsible × Role B	-	-0.760	0.575
Responsible × Role C	-	0.807	0.449
Responsible × Role D	-	-0.554	0.440
Responsible × Role E	-	0.613	0.440
Responsible × Role F	-	-0.693	0.439
Responsible × Role G	-	-0.474	0.477

Appendix Table A1: Full Results of the Bayesian Logistic Mixed-Effects Model

Notes: Dependent variable is the binary decision 'SWITCH'. Estimates represent posterior means from a Bayesian Logistic Mixed-Effects Model.
*** indicates |Post. Mean / Post. SD| > 1.96 (approximate 95% Bayesian Credible Interval does not contain zero).
Random Effects: LLM Architecture Variance $\sigma = 1.423$.

© Copyright National University of Singapore. All Rights Reserved.

Appendix

```
                               GEE Regression Results
==============================================================================
Dep. Variable:                decision_val   No. Observations:                 1680
Model:                                 GEE   No. clusters:                        3
Method:                        Generalized   Min. cluster size:                 560
                      Estimating Equations   Max. cluster size:                 560
Family:                           Binomial   Mean cluster size:               560.0
Dependence structure:         Independence   Num. iterations:                     2
Date:                     Wed, 01 Apr 2026   Scale:                           1.000
Covariance type:                    robust   Time:                         20:55:42
==============================================================================================
                                        coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                              3.1291      1.385      2.259      0.024       0.415       5.844
responsibility_name[T.Responsible]    -0.5812      0.538     -1.080      0.280      -1.636       0.473
role[T.role_B]                         2.7213      0.517      5.261      0.000       1.707       3.735
role[T.role_C]                        -3.3424      1.792     -1.865      0.062      -6.855       0.170
role[T.role_D]                        -1.6761      1.262     -1.328      0.184      -4.150       0.798
role[T.role_E]                        -1.3861      1.118     -1.240      0.215      -3.578       0.806
role[T.role_F]                        -1.8292      0.659     -2.774      0.006      -3.121      -0.537
role[T.role_G]                         0.1745      0.502      0.348      0.728      -0.808       1.158
model[T.doubao-seed-2-0-pro-260215]   -0.8645      0.165     -5.241      0.000      -1.188      -0.541
model[T.gemini-2.5-flash]             -3.7885      0.666     -5.685      0.000      -5.095      -2.482
=============================================================================
Skew:                          -0.3211   Kurtosis:                       0.5885
Centered skew:                 -0.3211   Centered kurtosis:              0.5885
=============================================================================
```

© Copyright National University of Singapore. All Rights Reserved.